Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3471-WE01

Title: (BEng/MEng)

Materials 3

Time Allowed:	2 hours
Additional Material provided:	Yes - Formula Sheet included as part of exam paper.
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Section A
Question 1

a) A steel alloy has gone through several cycles of strain hardening and annealing. The properties after each treatment are summarised in Table 1a. However for construction of a small gear, the yield strength of the steel is required to be 200 MPa. Calculate the maximum "average grain size" of the steel which meet the requirement of this application. What process do you recommend to achieve that?

Table 1a: Data related to Question 1a.

Average grain diamater (mm)	Yield strength (MPa)	Treatment
0.025	158	Cold working+ annealing
0.015	170	Cold working
0.050	145	Annealing
0.035	151	Hot working

[25%]

b) The kinetics of diffusion are strongly dependent on temperature, size of the diffusing atoms and the host material. Based on the information provided in Table 1b and Figure 1b, compare the ***D*** for nitrogen and hydrogen in iron at 950°C (1 bar) and explain the difference.
The ***D*** is Diffusion coefficient of the material and ***R***= 8.31 J/mol.K is the gas constant.

Table 1b: Diffusion data for iron (Question 1b).

Diffusion couple	Q (Joule/mol)	D_0 (cm^2/s)
N in FCC iron	144766	0.0034
N in BCC iron	76567	0.0047
H in FCC iron	43095	0.0063
H in BCC iron	15062	0.0012

[35%]

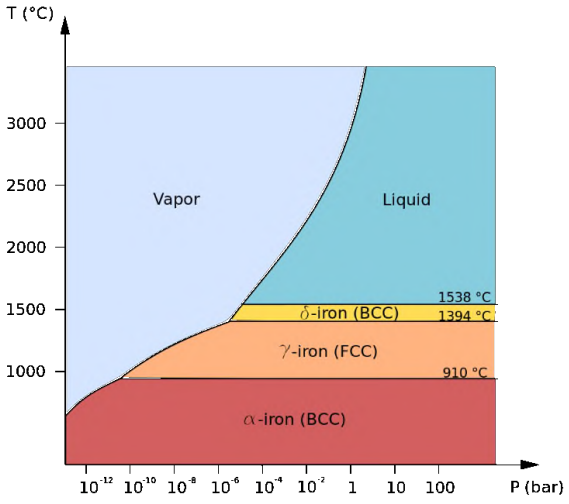


Figure 1b: phase diagram of pure iron (Question 1b)

continued

c) An unknown material's atomic radius is r = 1.75 Å. Determine the lattice constant of the materials if it is formed as (i) Simple cubic, (ii) BCC or (iii) FCC crystal structure.

[40%]

Section A
Question 2

a) An offshore wind farm's base features a steel nut with a steel bolt snugly fastened to it. After some time, the nut is found to have multiple cracks. Determine any likely reasons for cracking.

[15%]

b) For space application, an engineer created a fibre reinforced composite by incorporating 70% tungsten fibres into a Niobium matrix.The composite is composed of 70 vol% tungsten. What would happen if the composite is heated up to 2600°C (holding at that temperature for couple of months)? Justify your answer. (Refer to Figure 2a).
Denisty of Nb and W are 8.57 and 19.25 g/cm^3, respectively.

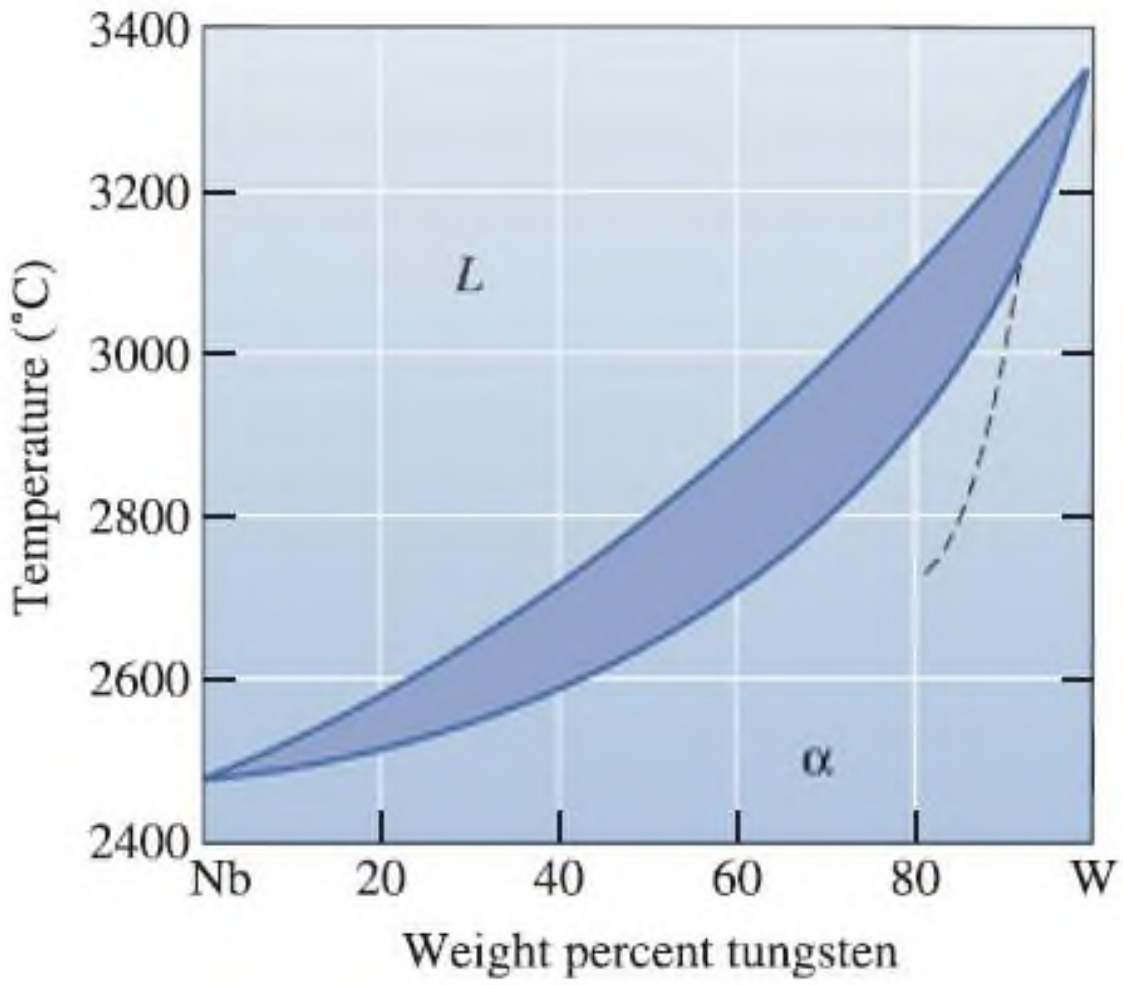


Figure 2b: The equilibrium phase diagram for the Nb-W system.
The dotted curve shows nonequilibrium solidus.

[50%]

continued

c) A PET rope is acting as a fastener in an intricate assembly and is under a stress of 550MPa. After 72 hours of constant strain, the stress falls to 540 MPa. Determine the life of the assembly if the stress on the part must remain above 512 MPa in order for the part to function properly.

[35%]

continued

Section B

Question 3

a) What is the definition of sustainable development?

[10%]

b) Explain what a Sustainable Development Goal by giving 3 examples and discuss in your own words or annotated diagrams how they interact and what role you think an engineer has in addressing them?

[60%]

c) Explain using diagrams and your own words why the UK Government wants to move from a linear economy to a circular economy and how this is relevant to sustainable development. Give one example of other piece of Government legislation which is encouraging this transition.

[30%]

continued

Section B

Question 4

Given the global interest in reducing CO_2 emission, the sales department of a company that manufactures the double decker London buses requested the engineering department in the same company to consider redesigning the chassis of the vehicle to ensure sustainable sales against their competitors. The chassis structural elements are highlighted in black in Figure Q4 – bus.

a) In the first meeting of the engineering department, A colleague presented a table of properties of materials (Table QT4) and proposed that the best choice would be the material with the lowest value in the "carbon footprint" column. Briefly explain if you agree or disagree with this proposal, and on either case justify your approach.

[25%]

b) Taking into consideration of the CO_2 associated with the production of the raw material, identify an appropriate material index. No need to derive it from first principles, but you should justify your methodology. Use the material index to rank the candidate materials. Take into consideration that making the vehicle lighter could provide an energy and CO_2 saving over the life of the vehicle due to reduced rolling resistance and energy to accelerate the vehicle. For a diesel bus on running relatively short distances with repeated stop-start service, the additional powertrain CO_2 output per km for each additional kg of vehicle mass can be taken to be 0.035gramsCO_2e / kg km. The design life of the truck is 1,000,000 km.

[25%]

c) What is the appropriate combination of material properties to be maximised to minimise the combined CO_2e considering the combination of production of the raw materials and the CO_2e emitted by the vehicle over its lifetime? Rank the candidate materials based on this revised metric.

[25%]

d) The company moves forward with the development of a new truck with a lighter chassis to provide a lifetime CO_2 saving. However, if bus operators decide to take the opportunity of the lighter chassis to increase the number of passenger that they can carry on each journey so that the loaded new truck remains as heavy as the old one. Does this undermine the benefit achieved by redesigning the chassis? Explain your rationale.

[25%]

continued

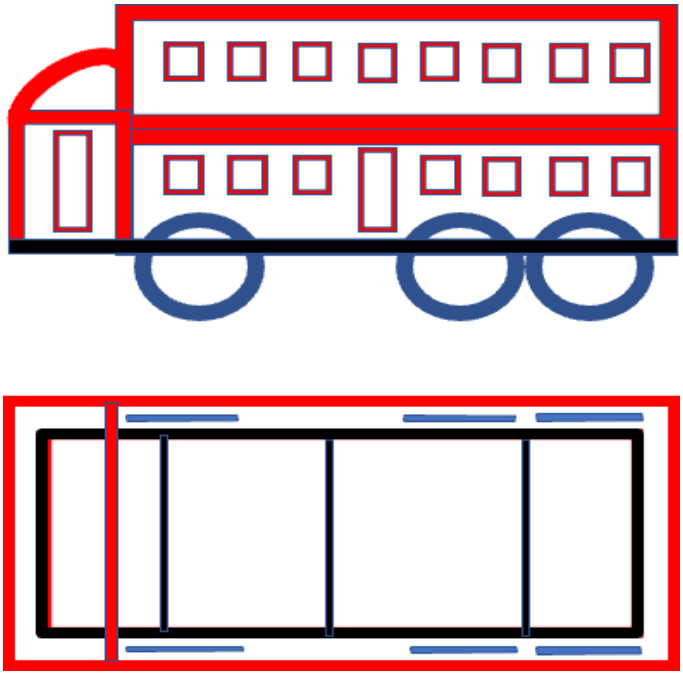


Figure 4 – Bus with chassis structural elements highlighted in black.

Material	E Young's Modulus (GPa)	σ_f Failure Stress (MPa)	ρ Density (kg/m^3)	C_F Carbon Footprint ($kgCO_{2e}/kg$)
Aluminium Alloy 6083T6	70.0	240	2700	12.5
Magnesium Alloy	44.5	220	1850	35.5
Steel (low carbon)	210	355	7850	1.9
Titanium Alloy 6AL4v	114	880	4430	46.5

Table 4: Material properties and indices.

continued

Section B

Question 5

a) An investigation is being carried out under laboratory conditions to determine the metal contained in a 200mm long weld base with a cross-sectional area of 8.0 mm^2. It is discovered that 4500J is required to melt the metal.

i) Using Table 5a, what is the metal most likely to be?

ii) If the heat transfer factor is 0.80 and the melting factor is 0.50 for a welding process, how much heat must be generated at the welding source to accomplish the weld?

Metal	Melting Temperature		Metal	Melting Temperature	
	°K[a]	°R[b]		°K[a]	°R[b]
Aluminium alloys	930	1680	Steels		
Cast iron	1530	2760	Low carbon	1760	3160
Copper and alloys			Medium carbon	1700	3060
Pure	1350	2440	High carbon	1650	2960
Brass, navy	1160	2090	Low alloy	1700	3060
Bronze (90Cu – 10 Sn)	1120	2010	Stainless steels		
Inconel	1660	3000	Austenitic	1670	3010
Magnesium	940	1700	Martensitic	1700	3060
Nickel	1720	3110	Titanium	2070	3730

[a] Kelvin scale = centigrade (Celsius) temperature + 273

[b] Rankine scale = Fahrenheit temperature + 460

Table 5a: Melting temperatures on the absolute temperature scale for selected metals

[30%]

b) MERP Engineering is a local SME specialising in welding operations. They have been asked to produce parts that involve welding together sheets of copper and steel each measuring 40cm x 30cm. In your opinion, what welding process should they use? Explain your reasoning in your own words and describe the issues MERP Engineering may encounter.

[40%]

c) 3D Element is a manufacturer providing metal powders for 3D printing applications. They receive an order from Engineeringclicks, a metal 3D printing manufacturer, to supply 20kg of nickel alloy powders in a month. In your opinion, which metal powder producing method should 3D Element use to make the nickel alloy powders. Explain your reasoning in your own words.

[30%]

continued

Formula sheet

Section A

		Metal	Electrode Potential E^0 (Volts)
Anodic	↑	$Li^+ + e^- \rightarrow Li$	−3.05
		$Mg^{2+} + 2e^- \rightarrow Mg$	−2.37
		$Al^{3+} + 3e^- \rightarrow Al$	−1.66
		$Ti^{2+} + 2e^- \rightarrow Ti$	−1.63
		$Mn^{2+} + 2e^- \rightarrow Mn$	−1.63
		$Zn^{2+} + 2e^- \rightarrow Zn$	−0.76
		$Cr^{3+} + 3e^- \rightarrow Cr$	−0.74
		$Fe^{2+} + 2e^- \rightarrow Fe$	−0.44
		$Ni^{2+} + 2e^- \rightarrow Ni$	−0.25
		$Sn^{2+} + 2e^- \rightarrow Sn$	−0.14
		$Pb^{2+} + 2e^- \rightarrow Pb$	−0.13
		$2H^+ + 2e^- \rightarrow H_2$	0.00 — (defined)
	↓	$Cu^{2+} + 2e^- \rightarrow Cu$	+0.34
		$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	+0.40
		$Ag^+ + e^- \rightarrow Ag$	+0.80
		$Pt^{4+} + 4e^- \rightarrow Pt$	+1.20
		$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	+1.23
Cathodic		$Au^{3+} + 3e^- \rightarrow Au$	+1.50

The standard reduction potentials for selected elements and reactions

Active, Anodic End	Magnesium and Mg alloys
	Zinc
	Galvanized steel
	5052 aluminum
	3003 aluminum
	1100 aluminum
	Alclad
	Cadmium
	2024 aluminum
	Low-carbon steel
	Cast iron
	50% Pb-50% Sn solder
	316 stainless steel (active)
	Lead
	Tin
	Cu-40% Zn brass
	Nickel-based alloys (active)
	Copper
	Cu-30% Ni alloy
	Nickel-based alloys (passive)
	Stainless steels (passive)
	Silver
	Titanium
	Graphite
	Gold
	Platinum
	Noble, Cathodic End

The galvanic series in seawater

continued

Bragg's Law:

$$sin\ \theta = \frac{\lambda}{2d_{hkl}}$$

Concentration of vacancies:

$$n_v = n \exp(-Q_v/RT)$$

ASTM grain size number:

$$N = 2^{n-1}$$

Percent cold work:

$$Percent\ of\ cold\ work = \left[\frac{A_0 - A_f}{A_0}\right] \times 100$$

Hall-Petch Equation:

$$\sigma_y = \sigma_0 + kd^{-1/2}$$

Diffusion coefficient dependency on Tempearture

$$D = D_0 \exp(\frac{-Q}{RT})$$

Stress relaxation:

$$\sigma = \sigma_0 \exp(-t/\lambda)$$

Relaxation time:

$$\lambda = \lambda_0 \exp(Q/RT)$$

Formula sheet

Section B

Screen mesh for sorting particle sizes

$$PS = \frac{1}{MC} - t_w$$

continued

Where PS = particle size, MC = mesh count, openings per linear mm and t_w = wire thickness of screen mesh.

Surface area (sphere)

$$A = \pi D^2$$

Where D = diameter of the spherical particle

Surface volume (sphere)

$$V = \frac{\pi D^3}{6}$$

Where D = diameter of the spherical particle

Area-to-volume ratio for a sphere

$$\frac{A}{V} = \frac{6}{D}$$

Where A = surface area, V = surface volume, D = diameter of the spherical particle

Area-to-volume ratio for any particle shape – spherical or non-spherical

$$\frac{A}{V} = \frac{K_S}{D} \text{ or } K_S = \frac{AD}{V}$$

Where K_S = shape factor; D = the diameter of a sphere of equivalent volume as the nonspherical particle.

continued

Power density

$$PD = \frac{P}{A}$$

where PD = power density, W/mm^2; P = power entering the surface, W; and A= surface area over which the energy is entering, mm^2.

Heat required to melt a given volume of metal

$$U_m = KT_m^2$$

where U_m = the unit energy for melting, J/mm^3; T_m = melting point of the metal on an absolute temperature scale, °K(°R); and K = constant whose value is 3.33 x 10^{-6} JK^{-2}mm^{-3}, when the Kelvin scale is used (and K = 1.467 x 10^{-5} JR^{-2}mm^{-3} for the Rankine temperature scale).

Heat energy available for welding:

$$H_w = f_1 f_2 H$$

where H_w = net heat available for welding, J, f_1 = heat transfer factor, f_2 = the melting factor, and H = the total heat generated by the welding process, J.

Energy balance equation

$$H_w = U_m V$$

Where H_w = net heat energy used by the welding operation, J; U_m = unit energy required to melt the metal, J/mm^3; and V = the volume of metal melted, mm^3.

Rate balance equation

$$R_{Hw} = U_m R_{WV}$$

Where R_{Hw} = rate of heat energy delivered to the operation for welding, J/s = W; and R_{WV} = volume rate of metal welded, mm^3/s.

$$R_{Hw} = f_1 f_2 R_H = U_m A_w v$$

here f_1 and f_2 are the heat transfer and melting factors; R_H = rate of input energy generated by the welding power source, W; A_w = weld cross-sectional area, mm^2; and v = the travel velocity of the welding operation, mm/s.

continued

Time to complete a single layer in rapid prototyping

$$T_i = \frac{A_i}{vD} + T_r$$

Where T_i= time to complete layer i; A_i = area of layer i; v = average scanning speed of the laser beam at the surface; D = diameter of the laser beam at the surface, and T_r = repositioning time between layers.

Build cycle time (stereolithography)

$$T_c = \sum_{i=1}^{nl} T_i$$

Where T_c = the STL build cycle time; and nl = the number of layers used to approximate the part.

END